R13

Code No: 117AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, September - 2021
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) What are the characteristics of a problem?
b) Explain about iterative deepening A* algorithm. [6+9]

2.a) Explain about Alpha-Beta Pruning.
b) Write short on bounded look- ahead strategy. [7+8]

3. What is propositional logic? Explain its syntax and logical connectivities with examples. [15]

4.a) Illustrate with the example that how the knowledge is represented by using semantic network.
b) When the extended semantic nets are useful? Explain with the example. [7+8]

5. Define Expert system. How expert systems differ from traditional system? With the neat sketch explain the architecture of expert system. [15]

6. Explain the following with illustrations:
a) Bayesian Belief Networks
b) Dempster-Shafer Theory. [7+8]

7. What is machine learning? How the data is classified using decision trees? [15]

8.a) How the knowledge is represented through semantic web?
b) Write about sentence analysis phases. [7+8]

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